CIE International A Level Physics

# Topic 11

# Particle Physics

Revision Booklet

**This booklet covers:**

- Atoms, Nuclei and Radiation
- Types of Radiation: $\alpha$, $\beta$ and $\gamma$
- Radioactive Decay Equations
- Fundamental Particles: Quarks
- Hadrons, Baryons and Mesons
- Leptons and Beta Decay

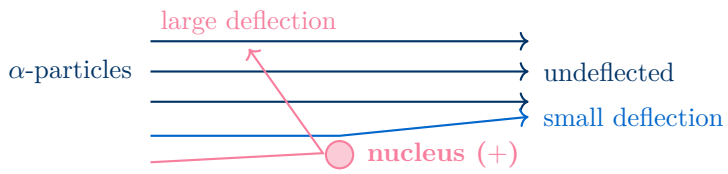
# Atoms, Nuclei and Radiation

## The Nuclear Atom

The **Geiger–Marsden** $\alpha$-particle scattering experiment (directed $\alpha$-particles at thin gold foil) showed:

- Most $\alpha$-particles passed straight through $\Rightarrow$ the atom is mostly empty space.
- A small fraction were deflected through large angles $\Rightarrow$ a dense, positively charged nucleus exists.
- A very few bounced back ($> 90°$) $\Rightarrow$ the nucleus is very small compared to the atom.

### Rutherford scattering diagram




## Nuclear Notation

A **nuclide** is a specific nuclear species characterised by its proton and nucleon numbers:

$$^{A}_{Z}\mathrm{X}$$

- $A$ = **nucleon number** (mass number): total number of protons + neutrons.
- $Z$ = **proton number** (atomic number): number of protons.
- Number of neutrons $N = A - Z$.
- **Isotopes**: atoms of the same element (same $Z$) with different numbers of neutrons (different $A$).

## Conservation Laws in Nuclear Processes

In every nuclear reaction or decay, the following are always conserved:

- **Nucleon number** $A$ (total count of protons + neutrons).
- **Charge** (equivalently, proton number $Z$).

These two conservation laws are the key tool for balancing decay equations.

## Unified Atomic Mass Unit

The **unified atomic mass unit** (u) is defined as exactly $\frac{1}{12}$ of the mass of a carbon-12 atom.

$$1\ \mathrm{u} = 1.661 \times 10^{-27}\ \mathrm{kg}$$

Approximate masses: proton $\approx 1$ u; neutron $\approx 1$ u; electron $\approx 0.00055$ u.

## Types of Radiation: $\alpha$, $\beta$ and $\gamma$

| Radiation | Composition | Charge | Mass | Stopped by |
|---|---|---|---|---|
| $\alpha$ (alpha) | 2 protons + 2 neutrons (helium-4 nucleus) | $+2e$ | 4 u | A few cm of air; thin paper |
| $\beta^-$ (beta-minus) | Electron ($e^-$) | $-e$ | $\approx 0$ ($\frac{1}{1836}$ u) | A few mm of aluminium |
| $\beta^+$ (beta-plus) | Positron ($e^+$) | $+e$ | $\approx 0$ ($\frac{1}{1836}$ u) | Annihilates with electron |
| $\gamma$ (gamma) | Electromagnetic wave / photon | 0 | 0 | Several cm of lead |

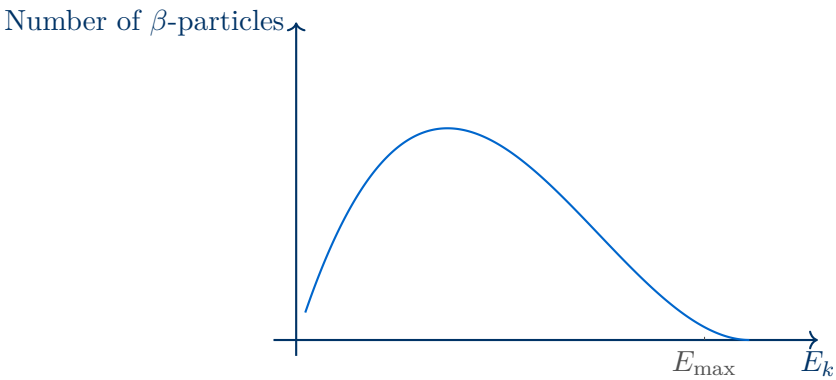
### Antiparticles

Every particle has a corresponding **antiparticle** with:

- The **same mass** as the particle.
- The **opposite charge** (and opposite quantum numbers).
- The **positron** ($\beta^+$, $e^+$) is the antiparticle of the electron ($e^-$).
- When a particle meets its antiparticle they **annihilate**, producing energy (usually two $\gamma$-ray photons).

### Energies of Emitted Particles

- **$\alpha$-particles** are emitted with **discrete (fixed) energies**: each decay of a given nuclide always releases an $\alpha$-particle of the same kinetic energy, giving a sharp line spectrum.
- **$\beta$-particles** have a **continuous range of energies** from zero up to a maximum: the total energy of the decay is shared between the $\beta$-particle and an (anti)neutrino, which can take any share — hence the continuous spectrum.
- **(Anti)neutrinos** are produced in $\beta$-decay to account for this energy sharing and to conserve momentum and angular momentum.




**Continuous $\beta$ energy spectrum**

**Common Mistake**

Students often state that $\beta$-particles have a fixed energy like $\alpha$-particles. They do not — the continuous spectrum is direct evidence for neutrino emission. The **maximum** $\beta$ energy equals the total energy available from the decay (as if no neutrino were present).

## Radioactive Decay Equations

### Alpha decay

In $\alpha$-decay, the nucleus emits a helium-4 nucleus ($^4_2\alpha$). Both $A$ and $Z$ decrease:

$$^A_Z\text{X} \longrightarrow {}^{A-4}_{Z-2}\text{Y} + {}^4_2\alpha$$

*Example:*

$$^{238}_{92}\text{U} \longrightarrow {}^{234}_{90}\text{Th} + {}^4_2\alpha$$

Check: $A$: $238 = 234 + 4$ ✓; $Z$: $92 = 90 + 2$ ✓

### Beta-minus decay ($\beta^-$)

A neutron converts to a proton; an electron and an electron antineutrino are emitted. $A$ is unchanged; $Z$ increases by 1:

$$^A_Z\text{X} \longrightarrow {}^{A}_{Z+1}\text{Y} + {}^0_{-1}\text{e} + \bar{\nu}_e$$

*Example:*

$$^{14}_{6}\text{C} \longrightarrow {}^{14}_{7}\text{N} + {}^0_{-1}\text{e} + \bar{\nu}_e$$

### Beta-plus decay ($\beta^+$)

A proton converts to a neutron; a positron and an electron neutrino are emitted. $A$ is unchanged; $Z$ decreases by 1:

$$^A_Z\text{X} \longrightarrow {}^{A}_{Z-1}\text{Y} + {}^0_{+1}\text{e} + \nu_e$$

### Gamma emission

$\gamma$-radiation accompanies $\alpha$ or $\beta$ decay when the daughter nucleus is left in an excited state. Neither $A$ nor $Z$ changes:

$$^A_Z\text{X}^* \longrightarrow {}^A_Z\text{X} + \gamma$$

### Decay Equation Summary

| Decay | $\Delta A$ | $\Delta Z$ | Particles emitted |
|---|---|---|---|
| $\alpha$ | $-4$ | $-2$ | ${}^{4}_{2}\alpha$ |
| $\beta^-$ | 0 | $+1$ | ${}^{0}_{-1}\mathrm{e}$, $\bar{\nu}_e$ |
| $\beta^+$ | 0 | $-1$ | ${}^{0}_{+1}\mathrm{e}$, $\nu_e$ |
| $\gamma$ | 0 | 0 | $\gamma$ photon |

## Fundamental Particles: Quarks

### Quarks

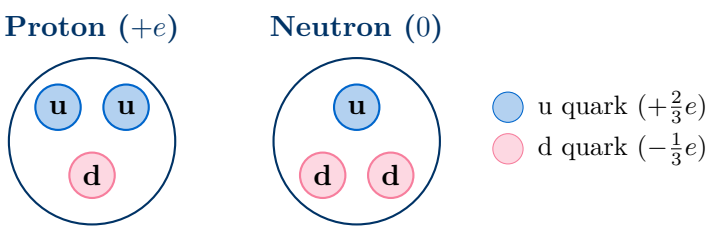
A **quark** is a fundamental, point-like particle. There are **six flavours** of quark (and six corresponding antiquarks):

| Flavour | Symbol | Charge | Antiquark | Symbol | Charge |
|---|---|---|---|---|---|
| up | u | $+\frac{2}{3}e$ | anti-up | $\bar{\mathrm{u}}$ | $-\frac{2}{3}e$ |
| down | d | $-\frac{1}{3}e$ | anti-down | $\bar{\mathrm{d}}$ | $+\frac{1}{3}e$ |
| strange | s | $-\frac{1}{3}e$ | anti-strange | $\bar{\mathrm{s}}$ | $+\frac{1}{3}e$ |
| charm | c | $+\frac{2}{3}e$ | anti-charm | $\bar{\mathrm{c}}$ | $-\frac{2}{3}e$ |
| top | t | $+\frac{2}{3}e$ | anti-top | $\bar{\mathrm{t}}$ | $-\frac{2}{3}e$ |
| bottom | b | $-\frac{1}{3}e$ | anti-bottom | $\bar{\mathrm{b}}$ | $+\frac{1}{3}e$ |

*Only up and down quarks are required for protons and neutrons; strange, charm, top and bottom are examined only for their charges.*

### Quark Composition of Proton and Neutron

| Particle | Quarks | Charge calculation | Total charge |
|---|---|---|---|
| Proton | uud | $\frac{2}{3}e + \frac{2}{3}e - \frac{1}{3}e$ | $+e$ |
| Neutron | udd | $\frac{2}{3}e - \frac{1}{3}e - \frac{1}{3}e$ | 0 |



## Hadrons, Baryons and Mesons

### Hadrons

A **hadron** is any particle made of quarks. Hadrons are subdivided into two classes:

- **Baryons**: composed of **three quarks** (qqq). Examples: proton (uud), neutron (udd).
- **Mesons**: composed of **one quark and one antiquark** ($q\bar{q}$). Examples: pion ($\pi^+ = u\bar{d}$).

Protons and neutrons are **not** fundamental particles — they are baryons made of quarks.

| Class | Quark content | Example | Charge |
|---|---|---|---|
| Baryon | qqq (three quarks) | Proton (uud) | $+e$ |
| Baryon | qqq (three quarks) | Neutron (udd) | 0 |
| Meson | $q\bar{q}$ (quark + antiquark) | $\pi^+$ ($u\bar{d}$) | $+e$ |
| Meson | $q\bar{q}$ (quark + antiquark) | $\pi^-$ ($\bar{u}d$) | $-e$ |
| Meson | $q\bar{q}$ (quark + antiquark) | $\pi^0$ ($u\bar{u}$ or $d\bar{d}$) | 0 |

## Leptons and Beta Decay

### Leptons

**Leptons** are fundamental (point-like, not made of quarks) particles. The leptons required at A-level are:

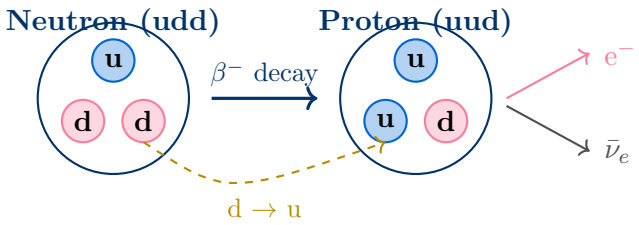
- **Electron** ($e^-$) — charge $-e$.
- **Positron** ($e^+$) — antiparticle of electron; charge $+e$.
- **Electron neutrino** ($\nu_e$) — charge 0; produced in $\beta^+$ decay.
- **Electron antineutrino** ($\bar{\nu}_e$) — charge 0; produced in $\beta^-$ decay.

### Quark Changes in Beta Decay

Beta decay occurs because a quark changes flavour inside a nucleon:

| Decay | Quark change | Nucleon change | Particles emitted |
|---|---|---|---|
| $\beta^-$ | $d \to u$ | Neutron $\to$ Proton | $e^- + \bar{\nu}_e$ |
| $\beta^+$ | $u \to d$ | Proton $\to$ Neutron | $e^+ + \nu_e$ |

### Quark-level picture of $\beta^-$ decay




**Neutrinos vs Antineutrinos**

This is a common source of error in exams:

- $\beta^-$ decay produces an **electron anti**neutrino ($\bar{\nu}_e$) — the antiparticle of the neutrino.
- $\beta^+$ decay produces an **electron neutrino** ($\nu_e$).

A helpful check: the lepton number must be conserved. In $\beta^-$, a lepton of number $+1$ (the electron) is created alongside a lepton of number $-1$ (the antineutrino), giving net lepton number zero — matching the original nucleon.

## Key Facts Summary

| Particle | Symbol | Charge | Type |
|---|---|---|---|
| Proton | p | $+e$ | Baryon (uud) |
| Neutron | n | 0 | Baryon (udd) |
| Electron | $e^-$ | $-e$ | Lepton |
| Positron | $e^+$ | $+e$ | Lepton (antiparticle of $e^-$) |
| Electron neutrino | $\nu_e$ | 0 | Lepton |
| Electron antineutrino | $\bar{\nu}_e$ | 0 | Lepton |
| $\alpha$-particle | ${}^4_2\alpha$ | $+2e$ | Baryon (not fundamental) |

**Quark charges (to recall):** u: $+\frac{2}{3}e$; d: $-\frac{1}{3}e$; s: $-\frac{1}{3}e$; c: $+\frac{2}{3}e$; t: $+\frac{2}{3}e$; b: $-\frac{1}{3}e$

**Antiquark charge = opposite of quark charge.**

**Conservation laws in every nuclear/particle process:** nucleon number $A$; charge $Z$; lepton number.

## Worked Examples

### Example 1 — Completing a Decay Equation

**Question:** Complete the following decay equation and identify the type of decay:

$$^{210}_{84}\text{Po} \longrightarrow {}^{A}_{Z}\text{X} + {}^{4}_{2}\alpha$$

**Solution**

Conserve nucleon number $A$: $210 = A + 4 \Rightarrow A = \mathbf{206}$
Conserve charge $Z$: $84 = Z + 2 \Rightarrow Z = \mathbf{82}$ (lead, Pb)

$$^{210}_{84}\text{Po} \longrightarrow {}^{206}_{82}\text{Pb} + {}^{4}_{2}\alpha$$

This is **alpha decay** — the nucleon number decreases by 4 and the proton number by 2.

### Example 2 — Identifying Decay Type from Equation

**Question:** Identify the particle X and the type of decay in:

$$^{90}_{38}\text{Sr} \longrightarrow {}^{90}_{39}\text{Y} + {}^{0}_{-1}\text{e} + X$$

**Solution**

$A$ is conserved: $90 = 90 + 0 + 0$ ✓
$Z$: $38 = 39 + (-1) + 0$ ✓
The emitted particles are an electron ($\beta^-$) and X. This is $\beta^-$ decay, so X must be an **electron antineutrino**: $X = \bar{\nu}_e$.

$$^{90}_{38}\text{Sr} \longrightarrow {}^{90}_{39}\text{Y} + {}^{0}_{-1}\text{e} + \bar{\nu}_e$$

### Example 3 — Quark Composition and Charge

**Question:** (a) State the quark composition of an antiproton. (b) Calculate its charge.

**Solution**

**(a)** The proton has quark composition uud. The antiproton is its antiparticle, so every quark is replaced by its antiquark:

$$\text{Antiproton} = \bar{\text{u}}\bar{\text{u}}\bar{\text{d}}$$

**(b)** Charge $= -\frac{2}{3}e + (-\frac{2}{3}e) + \frac{1}{3}e = -\frac{3}{3}e = \mathbf{-e}$
This confirms the antiproton has charge $-e$, equal in magnitude but opposite in sign to the proton.

### Example 4 — Quark Changes in $\beta^+$ Decay

**Question:** A proton undergoes $\beta^+$ decay. Write a full decay equation and describe the quark-level change.

**Solution**

$$^{1}_{1}\text{p} \longrightarrow {}^{1}_{0}\text{n} + {}^{0}_{+1}\text{e} + \nu_e$$

At the quark level, one **up quark converts to a down quark**:

$$\text{Proton (uud)} \longrightarrow \text{Neutron (udd)} + \text{e}^+ + \nu_e$$

Charge check: $+e \to 0 + e + 0$ ✓; Nucleon number: $1 = 1 + 0 + 0$ ✓

## Practice Exam Questions

### Section A — Short Answer Questions

**Q1.** Describe the key observations from the $\alpha$-particle scattering experiment and explain what each tells us about atomic structure.

*[4 marks]*

**Q2.** Distinguish between nucleon number and proton number. Define the term *isotope*.

*[3 marks]*

**Q3.** State the composition, charge and approximate mass (in u) of $\alpha$-, $\beta^-$- and $\gamma$-radiation.

*[5 marks]*

**Q4.** Explain why $\beta$-particles have a continuous energy spectrum, whereas $\alpha$-particles from a given nuclide have a fixed energy.

*[3 marks]*

## Section B — Longer Structured Questions

**Q5.** The nuclide $^{226}_{88}\text{Ra}$ undergoes $\alpha$-decay.

(a) Write a balanced nuclear equation for this decay, identifying the daughter nuclide.

*[2 marks]*

(b) State the two conservation laws used to balance the equation.

*[2 marks]*

(c) Explain why the $\alpha$-particles emitted in this decay all have the same kinetic energy.

*[2 marks]*

**Q6.** The nuclide $^{14}_{6}\text{C}$ undergoes $\beta^-$ decay.

(a) Write a full balanced equation for this decay, including all particles emitted.

*[2 marks]*

(b) Describe the quark-level change that occurs during this decay.

*[2 marks]*

(c) State and explain whether the $\beta^-$ particles emitted have a fixed or continuous range of energies.

*[2 marks]*

**Q7.** (a) State what is meant by a *hadron* and distinguish between a baryon and a meson.

*[3 marks]*

(b) The particle $\pi^+$ is a meson with quark composition $u\bar{d}$. Show that this gives a charge of $+e$.

*[2 marks]*

(c) State the quark composition of an antiproton and calculate its charge.

*[2 marks]*

(d) Explain why electrons are *not* classified as hadrons.

*[1 mark]*

## Mark Scheme and Answers

**Q1.** Most $\alpha$-particles pass straight through $\Rightarrow$ atom is mostly empty space [1]; small fraction deflected through large angles $\Rightarrow$ concentrated positive charge (nucleus) exists [1]; very few back-scattered $\Rightarrow$ nucleus is very small / very dense [1]; all deflections from positive charge are consistent with electrostatic repulsion [1].

**Q2. Nucleon number** ($A$): total number of protons + neutrons in a nucleus [1]; **proton number** ($Z$): number of protons only [1]; **isotopes**: atoms of the same element (same $Z$) with different numbers of neutrons (different $A$) [1].

**Q3.** $\alpha$: 2 protons + 2 neutrons; charge $+2e$; mass 4 u [2]. $\beta^-$: electron; charge $-e$; mass $\approx 0$ ($\frac{1}{1836}$ u) [2]. $\gamma$: electromagnetic radiation/photon; charge 0; mass 0 [1].

**Q4.** In $\beta$-decay, an (anti)neutrino is also emitted [1]; the total energy is shared between the $\beta$-particle and the (anti)neutrino in varying proportions [1]; so the $\beta$-particle can have any energy from zero to a maximum — continuous spectrum [1]. For $\alpha$: only two products (daughter + $\alpha$); by conservation of energy and momentum, the $\alpha$ always takes the same fixed share [implied/bonus].

**Q5(a).** ${}^{226}_{88}\text{Ra} \rightarrow {}^{222}_{86}\text{Rn} + {}^{4}_{2}\alpha$ [2] (1 mark if equation has correct concept but arithmetic error).

**Q5(b).** Conservation of nucleon number $A$ [1]; conservation of charge (proton number $Z$) [1].

**Q5(c).** Only two products (daughter + $\alpha$) [1]; by conservation of energy and momentum, the $\alpha$ always receives the same fixed kinetic energy (discrete energy levels) [1].

**Q6(a).** ${}^{14}_{6}\text{C} \rightarrow {}^{14}_{7}\text{N} + {}^{0}_{-1}\text{e} + \bar{\nu}_e$ [2].

**Q6(b).** A down quark converts to an up quark (d $\rightarrow$ u) [1]; inside a neutron, converting it to a proton [1].

**Q6(c).** Continuous range [1]; because the energy is shared between the $\beta^-$ particle and the antineutrino in varying proportions [1].

**Q7(a).** A hadron is a particle made of quarks [1]; a baryon consists of three quarks [1]; a meson consists of one quark and one antiquark [1].

**Q7(b).** Charge $= +\frac{2}{3}e + \frac{1}{3}e$ [1] $= +e$ ✓[1].

**Q7(c).** $\bar{\text{u}}\bar{\text{u}}\bar{\text{d}}$ [1]; charge $= -\frac{2}{3}e - \frac{2}{3}e + \frac{1}{3}e = -e$ [1].

**Q7(d).** Electrons are leptons, not composed of quarks / not hadrons by definition [1].

## Revision Checklist

Use this checklist to track your understanding. Tick each box when you are confident:

| | Learning Objective | Confidence (1–3) |
|---|---|---|
| ☐ | Describe the $\alpha$-scattering experiment and interpret its results | |
| ☐ | Describe the nuclear model: protons, neutrons and orbital electrons | |
| ☐ | Distinguish nucleon number ($A$) from proton number ($Z$) | |
| ☐ | Define isotopes and use the notation ${}^{A}_{Z}X$ | |
| ☐ | State that nucleon number and charge are conserved in nuclear processes | |
| ☐ | Describe composition, mass and charge of $\alpha$, $\beta^-$, $\beta^+$ and $\gamma$ | |
| ☐ | State that a positron is the antiparticle of an electron (same mass, opposite charge) | |
| ☐ | State that $\bar{\nu}_e$ is produced in $\beta^-$ decay and $\nu_e$ in $\beta^+$ decay | |
| ☐ | Explain why $\beta$-particles have a continuous energy spectrum | |
| ☐ | Write balanced $\alpha$- and $\beta$-decay equations | |
| ☐ | Use the unified atomic mass unit (u) | |
| ☐ | State the six quark flavours and recall their charges | |
| ☐ | State the quark composition of the proton (uud) and neutron (udd) | |
| ☐ | Distinguish hadrons (baryons: qqq; mesons: q$\bar{\text{q}}$) from leptons | |
| ☐ | Describe the quark change in $\beta^-$ (d→u) and $\beta^+$ (u→d) decay | |
| ☐ | Recall that electrons and neutrinos are leptons | |

*Key: 1 = Need more work 2 = Getting there 3 = Confident*

**Good luck with your revision!**

Particle physics is really just careful bookkeeping. Every decay equation is balanced by two rules — conserve $A$, conserve $Z$ — and every quark change in $\beta$-decay is just one quark swapping flavour. Get those two ideas solid and the whole topic clicks into place.